URCHIN – Underwater Robotic Construction Modules Using High-Tolerance Interlocking Nodes

Regular Paper

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Introduction

The increasing global demand for energy and natural resources has expanded the need for underwater infrastructure. Yet it remains resource-intensive and hazardous. Robotic automation can take divers out of the water, making construction safer and large-scale assembly practical [1], [2]. To address this, we propose URCHIN: a system of specialized voxel geometry that can mitigate the localization and manipulation errors inherent in marine environments. Specifically, this work demonstrates:

- **A self-aligning magnetic interface** backed by a magnetostatic model that predicts inter-voxel forces and informs design constraints, enabling autonomous construction without high-precision sensing.
- **A modular voxel design** featuring grasp-optimized geometry, making it compatible with general-purpose underwater manipulators.
- **A localization-ready design** with embedded marker slots for module tracking.

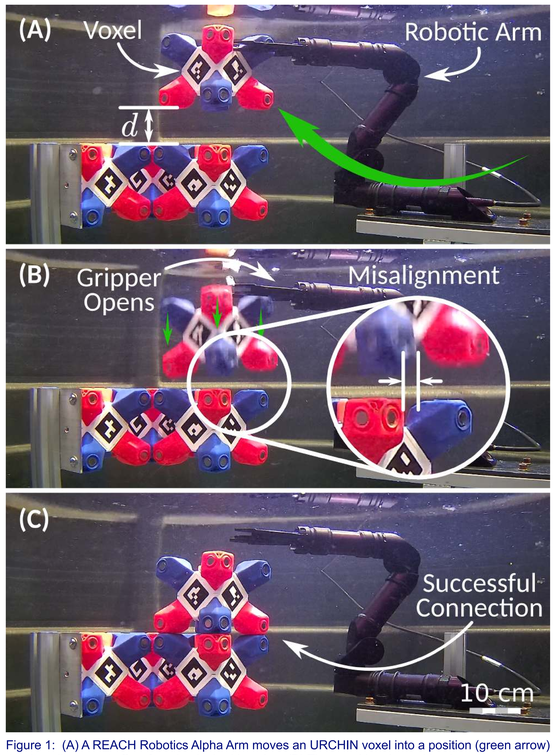


Figure 1: (A) A REACH Robotics Alpha Arm moves an URCHIN voxel into a position (green arrow) that is a distance d offset from its desired location. (B) The voxel is released by the gripper, and as it falls, it becomes misaligned. (C) The voxel automatically aligns and snaps into place.

Architecture

URCHIN is designed around the following:

- **Manipulator Compatibility:** The voxel geometry includes graspable legs with standardized spacing and flat surfaces tailored for common parallel-jaw grippers.
- **Passive Alignment:** Embedded neodymium magnets that facilitate automatic face-to-face alignment, allowing modules to snap together even under positional errors.
- **Module Tracking:** Integrated slots for localization markers.
- **Geometric Flexibility:** Support for multi-directional tiling and scaling to different sizes without altering core geometry.
- **Predictive Magnetic Modeling:** Magnetostatic interaction model that estimates inter-voxel forces during approach and alignment.

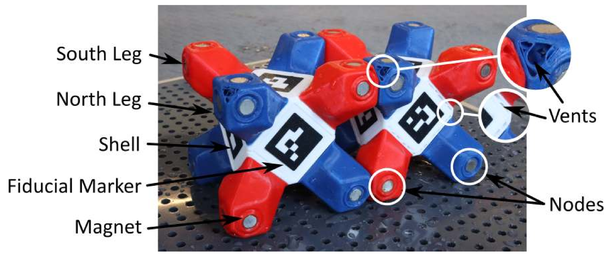


Figure 2: URCHIN voxel component overview and defining features.

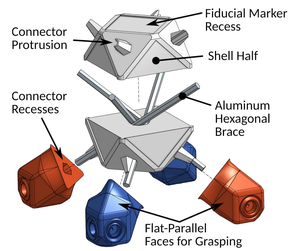


Figure 3: Exploded view of a voxel.

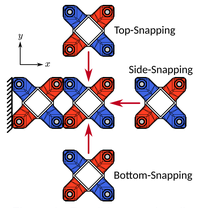


Figure 4: Possible snapping directions

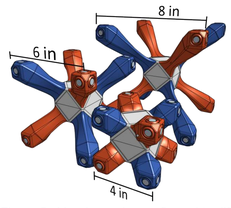


Figure 5: Voxels can be configured to different sizes.

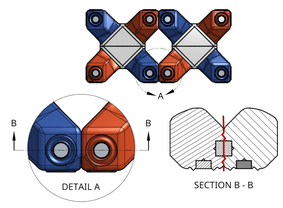


Figure 6: Section view of connection between north and south nodes.

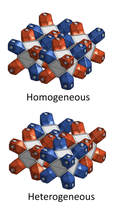


Figure 7: Voxels can have varying leg arrangements

Modeling

VOXEL-TO-VOXEL INTERACTION MODEL

To capture near-field ($\frac{1}{d^2}$) and far-field ($\frac{1}{d^4}$) behavior, we fit a reciprocal cubic model for inter-voxel pull force $F_{\text{est}}(d)$ as a function of separation distance d:

$$F_{\text{est}}(d) = \frac{aK}{bd^3 + cd^2 + ed + f} \tag{1}$$

where $K = \pi B^2 \frac{R^4}{\mu}$, with B representing the residual magnetic flux density (remanence), R the magnet pole radius, and μ the magnetic permeability of the medium (water). The empirical parameters (a, b, c, e, f) are solved via nonlinear optimization.

ANALYTICAL BENDING MODEL

The maximum supported external point force F_{applied} before joint detachment at the tip of a cantilevered chain of N voxels is given by:

$$F_{\text{applied}} \leq \frac{\frac{l}{L}F_{\text{ref}} - \frac{1}{2}N^2W}{N - \frac{1}{2}} \tag{2}$$

or equivalently expressed in moment form for the maximum allowable applied bending moment M_{applied} at the base joint:

$$M_{\text{applied}} \leq lF_{\text{ref}} - \frac{1}{2}N^2WL \tag{3}$$

where F_{ref} is the reference magnetic pull force at contact ($d = 0$), l is the rotation moment arm from the pivot edge to the magnetic axis, L is the characteristic voxel length, and W is the effective submerged weight of a single voxel.

Experiments & Results

SNAPPING DISTANCES

Evaluated Top, Side, and Bottom snapping underwater vs. land across offsets using a 4-DOF REACH Alpha Arm. Demonstrated reliable underwater snapping at offsets up to **4.44 cm (1.75 in)** top-down and **1.9 cm (0.75 in)** side-in.

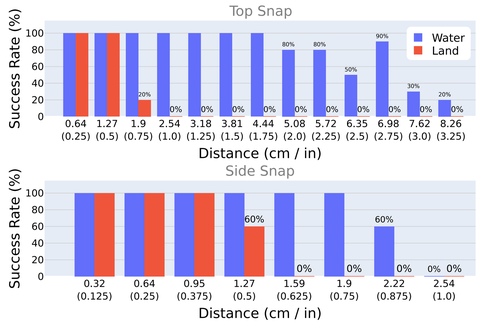


Figure 8: Snap success rates for top-down and side-in configurations, shown for both land and underwater conditions Each bar group represents 10 trials at a given offset.

PULL FORCE VERIFICATION

Measured axial face separation force using a Mark-10 tensile testing system with a 100N load cell. The reciprocal cubic model achieved an **RMSE of 0.118 N** (a **20× error reduction** over standard analytical Zurek models [3]).

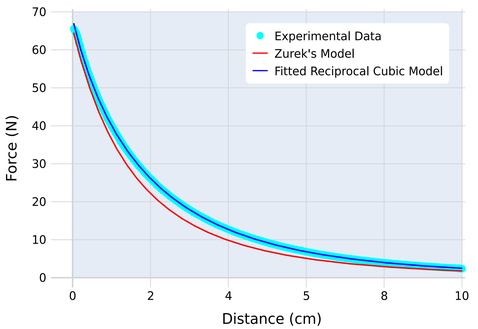


Figure 9: Measured magnet pull force between voxel faces compared to Zurek's model [3] and the fitted reciprocal cubic model as seen in Equation 1.

BENDING FAILURE

Evaluated cantilevered chains ($N = 1$ to 4 voxels) submerged in a water tank with an external pulley rig. Experimental detachment loads matched predictions by Equation 2 within an average error of **14.90%** across all chain lengths.

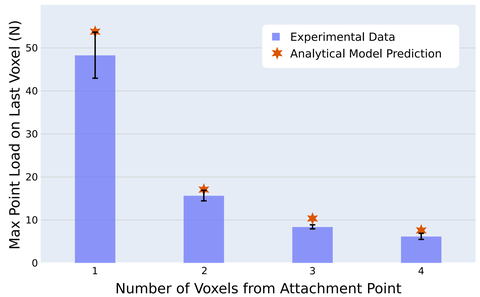


Figure 10: Maximum point load as a function of distance from the fixed end of the voxel chain.

AUTONOMOUS 1D/2D/3D ASSEMBLY

Executed unassisted placement in 1D, 2D corner, and 3D vertex configurations under forced initial misalignments. Demonstrated snapping success across multi-axis releases, successfully forming complex structures (arches, overhangs, branching chains).

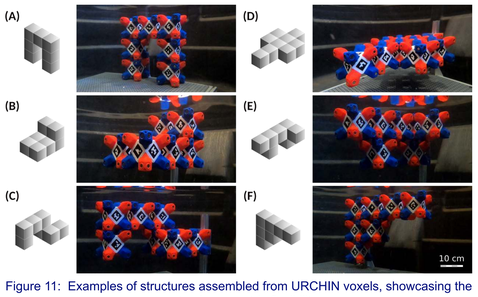


Figure 11: Examples of structures assembled from URCHIN voxels, showcasing the system's geometric flexibility.

Conclusion

Testing demonstrates promise for using magnets as an alignment interface for underwater construction. Future work includes the introduction of shear-resistant mechanism connector, and testing in dynamic ocean environments.

Bibliography

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[3] S. Zurek, "Performance of closed-form equations for force between cylindrical magnets over wide range of volume, aspect ratio, and force," *Journal of Electrical Engineering*, vol. 73, no. 6, pp. 405–412, Dec. 2022, doi: 10.2478/jee-2022-0055.

Acknowledgements

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